



Tennessee Valley Authority, Post Office Box 2000, Decatur, Alabama 35609-2000

October 31, 2017

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

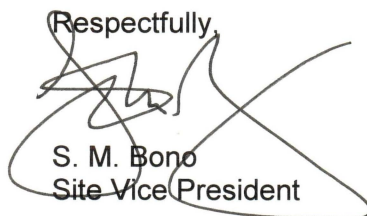
Browns Ferry Nuclear Plant, Unit 3
Renewed Facility Operating License No. DPR-68
NRC Docket No. 50-296

Subject: **Licensee Event Report 50-296/2017-001-00**

The enclosed Licensee Event Report provides details on the inoperability of a Residual Heat Removal Pump resulting in a condition prohibited by Technical Specifications. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73 (a)(2)(i)(B).

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact J. L. Paul, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,

A handwritten signature in black ink, appearing to read "S. M. Bono", is written over the typed name and title. The signature is stylized with a large, sweeping loop.

S. M. Bono
Site Vice President

Enclosure: Licensee Event Report 50-296/2017-001-00 – Inoperable Residual Heat Removal Pump Results in Condition Prohibited by Technical Specifications

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant

ENCLOSURE

**Browns Ferry Nuclear Plant
Unit 3**

Licensee Event Report 50-296/2017-001-00

Inoperable Residual Heat Removal Pump Results in Condition Prohibited by Technical Specifications

See Enclosed



LICENSEE EVENT REPORT (LER)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME
Browns Ferry Nuclear Plant, Unit 3

2. DOCKET NUMBER
05000296

3. PAGE
1 OF 8

4. TITLE
Inoperable Residual Heat Removal Pump Results in Condition Prohibited by Technical Specifications

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
09	01	2017	2017	- 001	- 00	10	31	2017	N/A	N/A
									FACILITY NAME	DOCKET NUMBER
									N/A	N/A

9. OPERATING MODE 1	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR 5: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
10. POWER LEVEL 100	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Mimi King-Patterson, Licensing Engineer

TELEPHONE NUMBER (Include Area Code)

(256) 614-7225

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX
B	BO	BKR	S125	Y	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE)☒ NO

15. EXPECTED SUBMISSION DATE

MONTH	DAY	YEAR
N/A	N/A	N/A

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On September 1, 2017, at approximately 1006 Central Daylight Time (CDT), Browns Ferry Nuclear Plant (BFN) Unit 3 3A Residual Heat Removal (RHR) system pump failed to start during performance of Surveillance 3-SR-3.5.1.6 (RHR I), Quarterly RHR System Rated Flow Test Loop I. The apparent cause was the Electrical Preventive Maintenance Instruction for 4kV Wyle/Siemens Horizontal Vacuum Circuit Breaker (Type-3AF) and Compartment Maintenance was revised to include steps to secure the breaker's mounting hardware which caused internal binding of the indication flag. Binding of the indication flag prevented the closing spring of the breaker from charging and the breaker from closing on demand. As a result, automatic start of the 3A RHR pump was prevented. On September 1, 2017, at approximately 1633 CDT, the 3A RHR Pump was declared operable following lubrication and testing of the breaker's indication flag mounting bolt.

A Past Operability Evaluation concluded that the 3A RHR Pump was inoperable from July 26, 2017 to September 1, 2017, which exceeded the Technical Specification allowed outage time. During this time, the 3B, 3C, and 3D RHR pumps would have started automatically upon receipt of an Emergency Core Cooling System (ECCS) initiation signal or from an Operator manual start demand from the Control Room. Based on results from the Probability Risk Assessment and Engineering inspections, there was no significant risk to the health and safety of the public or plant personnel for this event. The Corrective Action to reduce the probability of similar events occurring in the future will be addressed by revising the Electrical Preventive Maintenance Instruction for 4kV Wyle/Siemens breakers to ensure freedom of movement of the indication flag is present during the breaker inspection.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 3	05000-296	2017	- 001	- 00

NARRATIVE**I. Plant Operating Conditions Before the Event**

At the time of discovery, Browns Ferry Nuclear Plant (BFN), Unit 3 was in Mode 1 at 100 percent power.

II. Description of Event**A. Event Summary**

On September 1, 2017, at approximately 1006 Central Daylight Time (CDT), BFN Unit 3 3A Residual Heat Removal (RHR)[BO] system pump [P] failed to start during the performance of Surveillance, 3-SR-3.5.1.6 (RHR I), Quarterly RHR System Rated Flow Test Loop I. Maintenance troubleshooting revealed that the 3A RHR pump motor breaker's [BKR] closing spring failed to charge preventing the breaker from closing on demand. Malfunction of the 3A RHR pump motor breaker resulted in 3A RHR pump failing to start during manual or automatic actuation. Operations personnel declared 3A RHR pump inoperable.

A Past Operability Evaluation (POE) was performed for the 3A RHR pump failure. The POE concluded that the 3A RHR pump was inoperable from July 26, 2017 to September 1, 2017. BFN Technical Specifications (TS) Limiting Condition of Operability (LCO) 3.5.1 Required Action A.1 requires, with one required Emergency Core Cooling System (ECCS) injection/spray subsystem inoperable, that the required ECCS injection/ spray subsystem be restored to operable status within seven days. Additionally, BFN TS LCO 3.6.2.3 Required Action A.1 requires, with one RHR suppression pool cooling subsystem inoperable, restore the RHR suppression pool cooling subsystem to operable status within 30 days. The 3A RHR pump was inoperable for a time longer than allowed by TS. Therefore, The Tennessee Valley Authority (TVA) is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's TS.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

No inoperable systems, structures, or components contributed to this event.



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C. Dates and approximate times of occurrences

<u>Dates & Approximate Times</u>	<u>Occurrence</u>
July 26, 2017, 0859 CDT	3A RHR pump was started in Suppression Pool cooling mode to support Reactor Core Isolation Cooling (RCIC) [BN] flowrate surveillance testing.
July 26, 2017, 1110 CDT	3A RHR pump was removed from service following RCIC flowrate testing.
September 1, 2017, 1006 CDT	During performance of Surveillance 3-SR-3.5.1.6 (RHR I), 3A RHR pump failed to start. Operations personnel declared 3A RHR pump inoperable.
September 1, 2017, 1633 CDT	Operations personnel placed 3A RHR pump in service following Maintenance troubleshooting and post maintenance testing (PMT).

D. Manufacturer and model number of each component that failed during the event

The failed component during this event was the 3A RHR pump motor breaker. The model number for the 3A RHR pump motor breaker is 5-3AF-GEH-250-1200-58, manufactured by Siemens.

E. Other systems or secondary functions affected

There were no other systems or secondary functions affected by this condition.

F. Method of discovery of each component or system failure or procedural error

On September 1, 2017, during performance of Surveillance 3-SR-3.5.1.6 (RHR I), 3A RHR pump failed to start. Maintenance troubleshooting determined that the closing spring on 3A RHR pump motor breaker failed to charge due to internal binding of the breaker's indicating flag. The as-found condition of the breaker prevented the breaker from closing on demand. Due to the malfunction of 3A RHR pump motor breaker, the 3A RHR pump was unable to perform its design basis function.

G. The failure mode, mechanism, and effect of each failed component

This event was the result of internal binding of the 3A RHR pump motor breaker's charged/uncharged indication flag, which prevented the closing spring from charging and the breaker from closing on demand. With the breaker's actuating spring discharged, manual and automatic start of the 3A RHR pump was prevented.

It was discovered that although the 3A RHR pump motor breaker indicated charged and ready to close, the breaker actuating spring was actually discharged. Additional troubleshooting revealed that the breaker's charged indicating flag was bound, preventing actuation of the internal breaker

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charging limit switches. Since the limit switches could not change position following the last operation of the breaker, the ready to close light never extinguished and the charging motor never charged the breaker's actuating spring. This failure mechanism resulted in a false indication of the status of the breaker.

H. Operator actions

In response to indication that the 3A RHR pump was not functional, Operations personnel declared 3A RHR pump inoperable.

I. Automatically and manually initiated safety system responses

No safety system responses resulted from this event.

III. Cause of the event**A. Cause of each component or system failure or personnel error**

The Electrical Preventive Maintenance Instruction for 4kV Wyle/Siemens Horizontal Vacuum Circuit Breaker (Type-3AF) and Compartment Maintenance was revised on September 30, 2010 to include steps to verify that the breaker's spring charging motor mounting screws were secure. The procedure did not specify a step to inspect the indication flag mounting bolts. The mounting bolts are installed and inspected at the vendor's facility and are not normally adjusted during Preventive Maintenance (PM). Therefore, torque requirements were never provided for the indication mounting bolts. The inclusion of steps in the procedure to secure the spring charging motor mounting screws, without providing torque requirements, could have inadvertently caused overtightening of the indication flag. Therefore, any Wyle/Siemens breaker tested after September 30, 2010, would be more susceptible to binding of the indication flag. Thus, the apparent cause was the Electrical Preventive Maintenance Instruction for 4kV Wyle/Siemens Horizontal Vacuum Circuit Breaker (Type-3AF) and Compartment Maintenance provided a potential inaccurate step to secure all mounting hardware.

A 20 year old breaker test specimen was inspected at the vendor's facility. The vendor simulated the same failure mode of the breaker by tightening the indication flag mounting screw. When the mounting screw was tightened, it caused the indication flag to bind and become resistant to movement. However, loosening of the mounting screw allowed freedom of range of movement and the indication flag operated as designed.

The direct cause of this event was binding of the charged/uncharged indication flag, which prevented the closing spring from charging and the 3A RHR pump motor breaker from closing on demand. Malfunction of the breaker resulted in the 3A RHR pump failing to start during manual or automatic actuation.



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B. Cause(s) and circumstances for each human performance related root cause

There were no human performance root causes related to this event. All vendor guidance and maintenance procedures were followed. Although the apparent cause was an inaccurate step in the Electrical Preventive Maintenance Instruction for 4kV Wyle/Siemens Horizontal Vacuum Circuit Breaker (Type-3AF) and Compartment Maintenance to secure all mounting hardware, it was not reasonably foreseeable that slight tightening of the mounting bolts would cause binding of the indication flag. This step was included to preclude breaker inoperability due to loose bolts.

IV. Analysis of the event

This event resulted in BFN, Unit 3, 3A RHR Pump being inoperable for longer than allowed by the TS. With the 3A RHR pump motor breaker actuating spring discharged, the ability of the 3A RHR pump to automatically start or manually start was prevented.

The Safety objectives of the RHR System are as follows:

- a. To restore and maintain the coolant inventory in the reactor vessel so that the core is adequately cooled after a loss-of-coolant accident. The RHR System also provides cooling for the pressure suppression pool so that condensation of the steam resulting from the blowdown due to the design basis loss-of-coolant accident is ensured.
- b. To further extend the redundancy of the Core Standby Cooling Systems by providing for containment cooling.

The Low Pressure Coolant Injection (LPCI) subsystem is an integral part of the RHR System. It operates to restore and, if necessary, maintain the coolant inventory in the reactor vessel after a loss-of-coolant accident so that the core is sufficiently cooled. The LPCI mode of operation of the RHR System pumps water into the reactor vessel in time to flood the core and limit fuel clad temperature. During the period when 3A RHR pump was inoperable, 3B, 3C, and 3D RHR pumps were operable and available to perform this function.

A POE was performed for the 3A RHR pump. The POE concluded that the 3A RHR pump was inoperable from July 26, 2017 to September 1, 2017. BFN, Unit 3, TS LCO 3.5.1 requires that each ECCS injection/spray subsystem and the Automatic Depressurization System (ADS) function of six safety/relief valves shall be Operable in Mode 1, and Modes 2 and 3, except High Pressure Coolant Injection (HPCI) [BJ] and ADS valves are not required to be operable with reactor steam dome pressure less than or equal to 150 pounds per square inch gauge (psig). Required Action A.1 requires, with one required ECCS injection/spray subsystem inoperable, that the required ECCS injection/ spray subsystem be restored to operable status within seven days. Additionally, BFN TS LCO 3.6.2.3 requires that four RHR suppression pool cooling subsystems shall be operable. Required Action A.1 requires, with one RHR suppression pool cooling subsystem inoperable, restore the RHR suppression pool cooling subsystem to operable status within 30 days. The 3A RHR Pump, credited as a required low

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pressure ECCS Injection/spray and suppression pool cooling subsystem, was inoperable for a time longer than allowed by TS. Therefore, Unit 3, was in violation of TS 3.5.1 and 3.6.2.3 Required Action A.1.

V. Assessment of Safety Consequences

A Probabilistic Risk Assessment (PRA) was performed to evaluate the failure of the 3A RHR pump. The PRA concluded that the risk thresholds for the 3A RHR Pump was GREEN during the time of inoperability. The total change in Core Damage Frequency (CDF) and Large Early Release Frequency (LERF) for the failure of the 3A RHR pump from July 26, 2017 to September 1, 2017 was 3.06E-07 and 7.39E-08, respectively. The 1E-6 (CDF) and 1E-7 (LERF) thresholds for green were met.

There are a total of 93 safety related 4kV Wyle/Siemens breakers located in the plant. PMs were performed on 60 of the 93 breakers after the Electrical Preventive Maintenance Instruction for 4kV Wyle/Siemens Horizontal Vacuum Circuit Breaker (Type-3AF) and Compartment Maintenance was revised to include steps to secure the breaker's mounting hardware. These 60 breakers could be more susceptible to binding of the indication flag. Engineering performed a visual inspection of 43 out of the 60 safety related horizontal Wyle/Siemens breakers. These 43 breakers were chosen because they were in a non-protected status, were in the open position to preclude the potential trip of a closed breaker, and did not present a high safety risk to plant personnel. The inspection verified that the closing spring of the breakers were charged as expected. Work orders have been created to perform more intrusive inspections on all 60 breakers to test the indication flag. The 17 breakers that were excluded from the initial visual inspection will also be inspected to verify the closing springs are charged. These inspections will occur during planned outages to mitigate personnel safety risks and the potential of a plant trip or transient. While common mode failure in the remaining breakers is not expected, if the inspections result in identifying similar failures, a supplement will be provided to this LER.

Based on the above discussion, TVA has concluded that, during the time period that 3A RHR Pump was inoperable, there was no significant risk to the health and safety of the public or plant personnel for this event.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

During this event, RHR Pumps 3B, 3C, and 3D retained the ability to automatically start upon receipt of an ECCS Initiation signal or from an Operator manual start demand from the Control Room. Additionally, the 3A, 3B, 3C and 3D Core Spray pumps remained available and operable except during minimal periods where the core spray pumps were inoperable due to Surveillances being performed.

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B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

These conditions did not occur during a shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

This event resulted in inoperability of the 3A RHR pump for a time longer than allowed by TS, from the last time the 3A RHR Pump was started in Suppression Pool Cooling Mode to support RCIC Flowrate Surveillance testing on July 26, 2017, until the time of discovery of the condition on September 1, 2017.

VI. Corrective Actions

Corrective Actions (CAs) are being managed by TVA's corrective action program under Condition Report (CR) 1334534.

A. Immediate Corrective Actions

The indication flag mounting bolt was inspected, lubricated and tested. Engineering performed initial inspections of 43 risk significant, safety related horizontal Wyle/Siemens breakers to address common mode failure. The closing springs of the 43 breakers inspected were all charged as expected with no indication of common mode failure.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

The Corrective Action to reduce the probability of similar events occurring in the future will be addressed by revising the Electrical Preventive Maintenance Instruction for 4kV Wyle/Siemens Horizontal Vacuum Circuit Breaker (Type-3AF) and Compartment Maintenance to ensure freedom of movement of the indication flag is present during the Breaker inspection. If binding is present, adjustment of the indication flag mounting bolt will be made until freedom of movement is obtained.

Additional corrective actions include:

Inspections on all safety related Wyle/Siemens breakers will be performed to inspect freedom of movement of the indication flags and to ensure no other common mode failures exist.

A failure analysis of the breaker will be performed to determine cause of the 3A RHR Pump failure.

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VII. Previous Similar Events at the Same Site

A review of the BFN CAP and Licensee Event Reports (LERs) for Units 1, 2, and 3 revealed three similar events over the last four years:

LER 50-296/2014-003-00 describes a RHR Shutdown Cooling Inboard Suction Valve Isolation Relay [RLY] which was rendered inoperable for longer than allowed by TS due to improper landing of leads during PM. The event is similar because an improperly connected wire resulted in a long period of inoperability for a safety system. Corrective Actions were to develop and deliver a case study to the Maintenance, Modifications, and Operations departments based on the details of this event.

LER 50-260/2015-001-00 describes a failure of the 2A RHR Pump to manually start from the control room due to a loose fastener. The event is similar because an improperly latched hand switch resulted in a long period of inoperability for a safety system. Corrective Actions for this event were to discipline the individuals responsible, to tighten the loose fastener, and to revise maintenance instructions to reduce the probability of recurrence.

LER 50-296/2016-001-00 describes a failure of the 3A RHR Pump which was rendered inoperable for longer than allowed by TS due to failure of the 3A RHR Pump Motor Breaker Transfer Switch (MBTS) to fully latch due to binding. Binding of the MBTS resulted from being installed greater than its twenty-one year service life with no PM performed. Corrective Actions were to verify similar SB-1 transfer switches are latched in the NORMAL position on BFN, Units 1, 2, and 3, and to create a PM activity to periodically replace GE SB-1 transfer switches similar to the 3A RHR Pump MBTS on a 20 year frequency.

Corrective Actions from these LERs would not have prevented this event.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.